Finding the sweet spot via personalised Achilles tendon training: the future is within reach

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Musculoskeletal tissues, including tendon, are sensitive to their mechanical environment, with both excessive and insufficient loading resulting in reduced tissue strength. Although further research is required to elucidate the specific loading conditions (ie, magnitude, duration, frequency and rate) that maximise tendon adaptation, *in vitro*[1 2] and *in vivo*[3–6] evidence indicates tendon has an optimal range or 'sweet spot'[7] within which mechanical loading promotes positive tissue adaptation or homeostasis, with either too much or too little loading being detrimental (figure 1). *In vitro* bioreactor studies of rabbit Achilles tendon have shown that continuous 8 hours/day for 6 days of 6% cyclic strain at 0.25 Hz maintains[1] and promotes[2] tendon health, but the same duration and frequency with strain magnitude of 3% and 9% disrupted the extracellular matrix. Tendons that were load-deprived for 12 days increased their expression of collagen of inferior material properties (ie, type III) and had diminished mechanical performance. Health of these load-deprived tendons, represented by increased expression of type I collagen, was restored following the same loading regime (6 days, 8 hours/day, magnitude 6%, 0.25 Hz). *In vivo* studies have also systematically modulated Achilles tendon training regime by varying strain magnitude,[3 6 8] duration,[6] rate[6] and frequency.[3 6] Training at a frequency of 0.17 Hz and strain magnitude of 6.5% resulted in superior material properties (ie, increased stiffness) when compared with training with lower strain magnitude (~3%),[3] higher frequency (0.5 Hz),[3] higher rates (ie, one-legged jump)[6] or longer durations (12 s).[6] Conversely, tendon disuse, observed in elderly and spinal cord injury populations, has been associated with loss of stiffness.[4]

Whole-tendon deformation causes extracellular matrix (ECM) and tenocyte strains, which are believed to be important mechanical triggers for the biological cascade regulating structural adaptation. This cascade is affected by cell arrangement, shape and attachment to the ECM, as well as the local biochemical environment.[9] Computational model studies have further demonstrated that tendon strain and ECM damage can predict protein expression[9] and explain tendon adaptation.[10] Therefore, if it were possible to design personalised exercise programmes to strain tendon within the 'sweet spot', taking into account the current structure, material properties and biochemical environment, then clinical efficacy of exercise-based rehabilitation should improve.

Interindividual differences in musculotendinous structure, material properties and muscle activation patterns result in different internal tendon strain distributions when performing the same exercise, even when accounting for interindividual

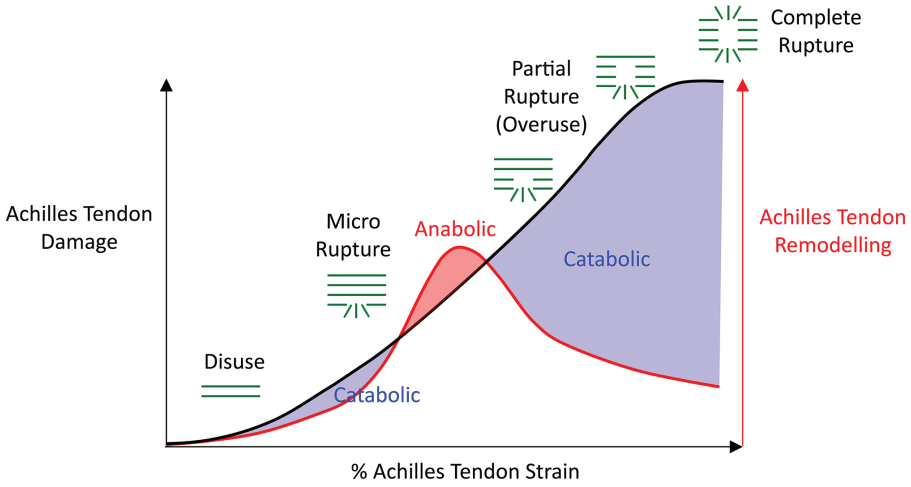

Figure 1 Schematic representation of the response of Achilles tendon to strain magnitude. Stimulating Achilles tendon within the 'sweet spot' results in positive (anabolic) adaptation. Conversely, underloading and overloading, due to disuse or overuse, respectively, results in detrimental (catabolic) response. Although not represented in this figure, strain frequency, strain duration and strain rate also affect Achilles tendon adaptation. Figure adapted from Wang *et al*[1] and Pizzolato *et al*.[7]

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differences in body weight and/or muscle strength. Standard exercise prescriptions such as a bout of eccentric heel drop exercise would therefore be expected to result in very different outcomes, even among individuals of the same age, gender, clinical state, training status and adherence to the programme. Unfortunately, directly measuring *in vivo* Achilles tendon strain during functional tasks, such as ambulation, is difficult. While two-dimensional ultrasound may be used to measure tendon strain non-invasively, these methods are prone to errors associated with probe movement and malalignment, especially during dynamic tasks.

Computational methods, such as musculoskeletal and finite element models, are a promising alternative to direct measurement of tissue strain and enable dynamic predictions of Achilles tendon strain.[7 11] Combined with inexpensive wearable technologies, these computational models can predict real-time Achilles tendon strain during functional tasks, thus enabling training to target the mechanobiological 'sweet spot'.[7] Such technology would connect, across size scales, knowledge from two previously distinct areas of science, that is, mechanobiology of isolated tendons and whole-body biomechanics, and usher in a new approach to Achilles tendon rehabilitation and training.

Accurate computational models of the Achilles tendon must be highly personalised to account for interindividual variations in tissue structure and material properties.[12] Achilles tendon structure can be reliably assessed during controlled isometric conditions using imaging three-dimensional ultrasound,[13] and a variety of experimental and computational methods to estimate tendon material properties are currently emerging. A recent finite element modelling study demonstrated Achilles tendon stresses during submaximal triceps surae contractions are highly sensitive to individual differences in tendon structure.[12] A future challenge will be to include heterogeneous and anisotropic material properties into such models to account for tendon lesions, structural discontinuities (eg, scar tissue) and tissue adaptation due to disuse or overuse. It will also be important to ensure the complex interplay between lower-limb kinematics, kinetics and muscle forces is accurately represented within the modelling framework. Sophisticated neuromusculoskeletal models that predict musculoskeletal tissue loading in real time from measured electromyograms,[14] as well as wearable sensors to directly measure tendon force,[15] have been developed and are currently being coupled to surrogate continuum models of the Achilles tendon to estimate localised tendon strain in real time. The aforementioned imaging methods, wearable sensors and computational models currently exist and have been validated independently, but require seamless integration before they can be applied in training and rehabilitation settings.

The main propositions of this editorial are that (1) the Achilles tendon can experience positive adaptation when exposed to strains within a specific range or 'sweet spot', (2) *in vivo* Achilles tendon strain depends on many factors known to differ between individuals and (3) new, integrated technologies that account for these interindividual differences and provide real-time feedback of Achilles tendon strain are within reach and will take some of the guesswork out of training and rehabilitation by ensuring tendon strains (magnitude, duration, frequency and rate) occur within the range that elicit maximal tendon adaptation. Further development of this technology could also help predict the complex local strain patterns experienced by tendons that contain tendinopathic lesions and/or structural discontinuities,[11] and thereby facilitate tendon regeneration through targeted training. These new technologies are just a few years from reality and have potential to change the way we approach rehabilitation and training of the Achilles tendon and musculoskeletal tissues in general.

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